



Weak wall boundary conditions for compressible flows

Monu Jaiswal¹ · Manoj R. Rajanna² · Md. Rhyhanul Islam¹ · Ming-Chen Hsu¹ · Yuri Bazilevs^{3,4}

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Abstract

Weak imposition of essential boundary conditions (i.e., weak BCs) for the Navier–Stokes equations of incompressible flows allows a certain amount of controlled numerical flow slip on the solid surface. Numerical flow slip mimics the presence of a thin boundary layer that would otherwise need to be captured using a fine mesh resolution. As a result, weak BCs enable the use of coarser meshes near solid walls without sacrificing numerical solution accuracy, which significantly reduces the computational costs, especially for 3D, wall-bounded turbulent flows. However, weak BCs for compressible flows are not as well understood as those for the incompressible-flow case. In particular, numerical instabilities were observed in some cases where the weak BCs were simultaneously imposed for the velocity and temperature fields. In the present effort, to address these stability issues, we develop a methodology for the design of compressible-flow weak BC operators and demonstrate the improved performance of the resulting weak BC formulations using challenging 2D and 3D test cases.

Keywords Compressible flow · Stabilized method · Weak boundary conditions · Near-wall modeling · Airfoil

1 Introduction

A major contributor to the computational cost in flow modeling is the presence of thin velocity and temperature boundary layers that develop near solid surfaces. Accurately resolving these layers typically requires highly refined meshes in the wall-normal direction. This results in a large number of elements, which leads to a high number of degrees of freedom (DoFs). Moreover, such refinement often introduces high-aspect-ratio elements, which can affect the solver's

numerical stability. Resolving these boundary layers is especially difficult in flow simulations with high Reynolds numbers or in three-dimensional settings, where the computational demand increases rapidly with mesh refinement.

In the context of stabilized and multiscale methods in fluid mechanics [1–3], weakly enforced essential boundary conditions (i.e., weak BCs [4]) have been introduced to mitigate the cost of fully resolving boundary layers. The key idea is to avoid imposing wall boundary conditions directly on boundary mesh nodes (or control points, as in Isogeometric Analysis (IGA) [5]). Instead, the boundary conditions are applied in a weak or variational manner that augments the flow formulation with appropriately designed boundary terms. This weak enforcement effectively relaxes strict no-slip conditions by allowing the flow to numerically slip at the wall, with the amount of slip controlled by the mesh resolution, local velocity, and viscosity. As a result, the discretization need not explicitly resolve the sharp gradients within boundary layers. This enables the use of coarser meshes near the wall, which significantly reduces computational cost while maintaining accurate predictions of global flow behavior. Numerical studies have shown that weak BC formulations can achieve accuracy comparable to or even surpassing that of strongly enforced methods, with significantly fewer DoFs [6–10].

✉ Ming-Chen Hsu
jmchsu@iastate.edu

✉ Yuri Bazilevs
yuri_bazilevs@brown.edu

¹ Department of Mechanical Engineering, Iowa State University, Ames, IA 50011, USA

² Global Engineering and Materials, Inc., Princeton, NJ 08540, USA

³ School of Engineering, Brown University, Providence, RI 02912, USA

⁴ Faculty of Science and Engineering, Graduate School of Creative Science and Engineering, Waseda University, Shinjuku, Tokyo 169-8050, Japan

The weak BCs for incompressible flows have been closely linked to near-wall modeling techniques, where the formulations resemble traditional wall functions and have been reparameterized to incorporate algebraic “law of the wall” relations for wall-bounded turbulent flows [11, 12]. The weak BC approach has also been successfully applied to immersogeometric analysis (IMGA) [13], where no-slip boundary conditions are imposed weakly on the immersed surface, leading to a highly effective approach for non-boundary-fitted meshes [14–22]. More recently, weak BC formulations have been extended to shifted boundary methods (SBM) for incompressible flows to impose boundary conditions on geometrically shifted interfaces [23, 24]. These works have established weak BCs as a unifying and versatile approach for boundary-condition enforcement in wall-bounded flows across various numerical frameworks.

However, weak BCs for compressible flows are not as well understood as their incompressible-flow counterparts. The first weak BC formulation for compressible flows was introduced by [25]. The main idea in [25] was to first develop a stabilized compressible-flow formulation using pressure-primitive variables, which has structural similarities to the incompressible-flow case [26]. These similarities enabled the authors to: (1) adapt the incompressible-flow weak velocity BC operator to the compressible-flow case with minimal modifications, and (2) add a weak temperature BC as a standalone operator derived from the advection–diffusion structure of the temperature equation. The resulting formulation performed reasonably well (see, e.g., [25, 27]), but instabilities were observed in some cases. In particular, for computations in the hypersonic regime reported in [28], stable weak enforcement of the velocity BCs at the solid wall could only be achieved if the temperature BCs at the wall were prescribed strongly. In addition, stable enforcement of the heat-flux BCs for the transonic flow cases reported in [29] could only be achieved if the normal component of the velocity vector at the solid wall was enforced strongly. In other words, fully weak enforcement of the wall velocity condition led to numerical instabilities in the compressible-flow regime. These observations underscore the need to improve existing weak BC formulations for compressible flows. Addressing these limitations and developing a robust, fully weak BC framework for compressible-flow simulation is the main focus of the present contribution.

The present paper is organized as follows. In Sect. 2, using the Navier–Stokes equations of compressible flows written in pressure-primitive variables and a framework of advective–diffusive systems, we present our methodology for developing weak wall BC operators for temperature and heat-flux boundary conditions. In Sect. 3, we compute challenging 2D and 3D aerodynamic test cases using the proposed weak BC formulations to, in part, demonstrate their

superior performance relative to the strong BC enforcement. Comparisons with experimental data are shown where such data is available. In Sect. 4, we draw conclusions and provide an outlook on using our weak BC formulations to develop a new class of immersed methods for compressible flows and fluid–structure interaction (FSI).

2 Methodology

This section develops a new formulation for the weak enforcement of wall boundary conditions for the Navier–Stokes equations of compressible flows. All equations are written in the Eulerian frame. Throughout the paper, $(\cdot)_{,t}$ denotes a partial time derivative taken with respect to a fixed spatial coordinate and $(\cdot)_{,i}$ denotes a spatial gradient, where $i = 1, \dots, d$, and $d = 2, 3$ is the space dimension. Einstein’s summation convention for repeated indices is used throughout.

2.1 Strong form

The reduced form of the Navier–Stokes equations of compressible flows from [25, 27] is taken as a starting point of the developments that follow. We introduce the vectors of conservation ($\mathbf{U}$) and pressure-primitive ($\mathbf{Y}$) variables as

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u_1 \\ \rho u_2 \\ \rho u_3 \\ \rho e \end{bmatrix} \quad \text{and} \quad \mathbf{Y} = \begin{bmatrix} p \\ u_1 \\ u_2 \\ u_3 \\ T \end{bmatrix}, \quad (1)$$

where the former is in a reduced form. In the above, ρ is the density, u_i is the i^{th} velocity component, p is the pressure, and T is the temperature. In addition, $e = c_v T$ is the fluid internal energy density, where $c_v = R/(\gamma - 1)$ is the specific heat at constant volume, R is the ideal gas constant, and γ is the heat capacity ratio. Finally, the pressure, density, and temperature unknowns are related through the ideal gas equation of state, $p = \rho R T$.

Using conservation variables, the Navier–Stokes equations of compressible flows hold in a fixed spatial domain $\Omega \in \mathbb{R}^d$ and may be written as

$$\mathbf{U}_{,t} + \mathbf{F}_{i,i}^{\text{adv}\setminus p} + \mathbf{F}_{i,i}^p + \mathbf{F}^{\text{sp}} - \mathbf{F}_{i,i}^{\text{diff}} - \mathbf{S} = \mathbf{0}, \quad (2)$$

where the convective flux (without the pressure) is given by

$$\mathbf{F}_i^{\text{adv}\setminus p} = \begin{bmatrix} \rho u_i \\ \rho u_i u_1 \\ \rho u_i u_2 \\ \rho u_i u_3 \\ \rho u_i e \end{bmatrix}, \quad (3)$$

the pressure flux is given by

$$\mathbf{F}_i^p = \begin{bmatrix} 0 \\ p\delta_{1i} \\ p\delta_{2i} \\ p\delta_{3i} \\ 0 \end{bmatrix}, \quad (4)$$

the diffusive flux is given by

$$\mathbf{F}_i^{\text{diff}} = \begin{bmatrix} 0 \\ \tau_{1i} \\ \tau_{2i} \\ \tau_{3i} \\ -\phi_i^q \end{bmatrix}, \quad (5)$$

the contribution of stress-power in the energy equation is given by

$$\mathbf{F}_i^{\text{sp}} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ pu_{i,i} - \tau_{ij}u_{j,i} \end{bmatrix}, \quad (6)$$

and $\mathbf{S}$ is the source term. In Eqs. (4)–(6), δ_{ij} is the Kronecker delta, and τ_{ij} and ϕ_i^q are the viscous stress and heat flux, respectively, given by $\tau_{ij} = \lambda u_{k,k}\delta_{ij} + \mu(u_{i,j} + u_{j,i})$ and $\phi_i^q = -\kappa T_{,i}$, where μ is the dynamic viscosity, $\lambda = -2\mu/3$, and κ is the thermal conductivity.

2.2 Weak forms

A weak form of the compressible-flow equations may be stated using conservation variables as follows: Find $\mathbf{U}$, such that for all vector-valued test functions $\mathbf{W}$,

$$\begin{aligned} & \int_{\Omega} \mathbf{W} \cdot \mathbf{U}_{,t} \, d\Omega + \int_{\Omega} \mathbf{W} \cdot \mathbf{F}_{i,i}^{\text{adv}\backslash p} \, d\Omega - \int_{\Omega} \mathbf{W}_{,i} \cdot \mathbf{F}_i^p \, d\Omega \\ & + \int_{\Omega} \mathbf{W}_{,i} \cdot \mathbf{F}_i^{\text{diff}} \, d\Omega + \int_{\Omega} \mathbf{W} \cdot \mathbf{F}^{\text{sp}} \, d\Omega \\ & - \int_{\Omega} \mathbf{W} \cdot \mathbf{S} \, d\Omega - \int_{\Gamma_H} \mathbf{W} \cdot \mathbf{H} \, d\Gamma = 0, \end{aligned} \quad (7)$$

where $\Gamma_H \in \mathbb{R}^d$ is part of the boundary of Ω with prescribed traction and heat-flux boundary conditions, with their value recorded in the vector $\mathbf{H}$. Note that the pressure and diffusive fluxes are integrated by parts in the above formulation to enable natural imposition of traction and heat-flux boundary conditions.

While a popular choice, conservation variables are perhaps not the most convenient option for computation (see, e.g., a discussion in [30]). A more natural choice is presented by the pressure-primitive variables. To reformulate the weak form of the compressible-flow problem given by

Eq. (7) using pressure-primitive variables, we first use the chain rule of differentiation to compute:

$$\mathbf{U}_{,t} = \mathbf{U}_{,\mathbf{Y}} \mathbf{Y}_{,t} = \mathbf{A}_0 \mathbf{Y}_{,t} \quad (8)$$

and

$$\mathbf{F}_{i,i}^{\text{adv}\backslash p} = \mathbf{F}_{i,\mathbf{Y}}^{\text{adv}\backslash p} \mathbf{Y}_{,i} = \mathbf{A}_i^{\text{adv}\backslash p} \mathbf{Y}_{,i}. \quad (9)$$

In addition, a direct computation gives

$$\mathbf{F}_i^p = \mathbf{A}_i^p \mathbf{Y}, \quad (10)$$

and

$$\mathbf{F}_{i,i}^{\text{diff}} = (\mathbf{K}_{ij} \mathbf{Y}_{,j})_{,i}. \quad (11)$$

Introducing the above relations into the weak form given by Eq. (7), we arrive at the weak form for the pressure-primitive variables: Find $\mathbf{Y} = [p, \mathbf{u}, T]^T$, such that for all vector-valued test functions $\mathbf{W} = [\delta p, \delta \mathbf{u}, \delta T]^T$,

$$\begin{aligned} & \int_{\Omega} \mathbf{W} \cdot \mathbf{A}_0 \mathbf{Y}_{,t} \, d\Omega \\ & + \int_{\Omega} \mathbf{W} \cdot \mathbf{A}_i^{\text{adv}\backslash p} \mathbf{Y}_{,i} \, d\Omega - \int_{\Omega} \mathbf{W}_{,i} \cdot \mathbf{A}_i^p \mathbf{Y} \, d\Omega \\ & + \int_{\Omega} \mathbf{W}_{,i} \cdot \mathbf{K}_{ij} \mathbf{Y}_{,j} \, d\Omega + \int_{\Omega} \mathbf{W} \cdot \mathbf{F}^{\text{sp}} \, d\Omega \\ & - \int_{\Omega} \mathbf{W} \cdot \mathbf{S} \, d\Omega - \int_{\Gamma_H} \mathbf{W} \cdot \mathbf{H} \, d\Gamma = 0, \end{aligned} \quad (12)$$

Detailed expressions for the matrices $\mathbf{A}_0$, $\mathbf{A}_i^{\text{adv}\backslash p}$, $\mathbf{A}_i^p$, and $\mathbf{K}_{ij}$ appearing in Eq. (12) may be found in [25, Appendix A].

2.3 Design of weak wall BC operators

To design a weak wall boundary condition (BC) operator, active on the wall boundary denoted by Γ_G , we first focus on the convective and pressure flux terms. We first note that the convective-flux terms are not integrated by parts, while the pressure-flux terms are. For this, the following terms are added to Eq. (12) or Eq. (7):

$$\begin{aligned} & - \int_{\Gamma_G} \mathbf{W} \cdot \left(\mathbf{F}_i^{\text{adv}\backslash p}(\mathbf{Y}) - \mathbf{F}_i^{\text{adv}\backslash p}(\mathbf{G}) \right) n_i \, d\Gamma \\ & + \int_{\Gamma_G} \mathbf{W} \cdot \mathbf{F}_i^p(\mathbf{G}) n_i \, d\Gamma. \end{aligned} \quad (13)$$

Here $\mathbf{G}$ denotes a vector of prescribed wall BC values and n_i is the i^{th} component of the unit outward normal vector.

For the case of pressure-primitive variables, $\mathbf{G}$ contains the known values of $[p, \mathbf{u}, T]^T$. (We will consider specific cases in the following sections.) We examine the first term in Eq. (13), expand the difference in Taylor series and only keep the linear terms as

$$\begin{aligned} \mathbf{F}_i^{\text{adv}\backslash p}(\mathbf{Y}) &\approx \mathbf{F}_i^{\text{adv}\backslash p}(\mathbf{G}) + \mathbf{F}_{i,\mathbf{Y}}^{\text{adv}\backslash p}|_{\mathbf{G}} (\mathbf{Y} - \mathbf{G}) \\ &= \mathbf{F}_i^{\text{adv}\backslash p}(\mathbf{G}) + \mathbf{A}_i^{\text{adv}\backslash p}|_{\mathbf{G}} (\mathbf{Y} - \mathbf{G}). \end{aligned} \quad (14)$$

Introducing the above outcome together with Eq. (10) into Eq. (13) results in the weak wall BC operator given by

$$\begin{aligned} & - \int_{\Gamma_G} \mathbf{W} \cdot [\mathbf{A}_i^{\text{adv}\backslash p}|_{\mathbf{G}} n_i] (\mathbf{Y} - \mathbf{G}) \, d\Gamma \\ & + \int_{\Gamma_G} \mathbf{W} \cdot [\mathbf{A}_i^p n_i] \mathbf{G} \, d\Gamma. \end{aligned} \quad (15)$$

We now examine two wall BC cases corresponding to a prescribed temperature or heat flux. In the case of constant temperature, the BC is called isothermal, and in the case of zero heat flux, it is referred to as adiabatic. In both cases, the flow velocity at the wall is zero. Pressure and density are unknown at the wall and are computed from an equation of state (the ideal gas law is assumed in this work).

2.3.1 Temperature wall BCs

For the temperature wall BC, the vector $\mathbf{G}$ is given as follows

$$\mathbf{G} = \begin{bmatrix} p \\ 0 \\ 0 \\ 0 \\ T_g \end{bmatrix}, \quad (16)$$

where T_g is the prescribed temperature (possibly a spatially and temporally varying function), zeros in the middle reflect zero flow velocity, and p is an unknown taken from the solution. With these values, the matrices $\mathbf{A}_i^{\text{adv}\backslash p}|_{\mathbf{G}} n_i$ and $\mathbf{A}_i^p n_i$ become (see [25, Appendix A]):

$$\mathbf{A}_i^{\text{adv}\backslash p}|_{\mathbf{G}} n_i = \begin{bmatrix} 0 & \rho n_1 & \rho n_2 & \rho n_3 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & \rho c_v T_g n_1 & \rho c_v T_g n_2 & \rho c_v T_g n_3 & 0 \end{bmatrix} \quad (17)$$

and

$$\mathbf{A}_i^p n_i = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ n_1 & 0 & 0 & 0 & 0 \\ n_2 & 0 & 0 & 0 & 0 \\ n_3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \quad (18)$$

The products $[\mathbf{A}_i^{\text{adv}\backslash p}|_{\mathbf{G}} n_i] (\mathbf{Y} - \mathbf{G})$ and $[\mathbf{A}_i^p n_i] \mathbf{G}$ are obtained from a direct computation as

$$[\mathbf{A}_i^{\text{adv}\backslash p}|_{\mathbf{G}} n_i] (\mathbf{Y} - \mathbf{G}) = \begin{bmatrix} \rho (u_i n_i - 0) \\ 0 \\ 0 \\ 0 \\ \rho c_v T_g (u_i n_i - 0) \end{bmatrix} \quad (19)$$

and

$$[\mathbf{A}_i^p n_i] \mathbf{G} = \begin{bmatrix} 0 \\ p n_1 \\ p n_2 \\ p n_3 \\ 0 \end{bmatrix}. \quad (20)$$

Introducing the above results in Eq. (15) gives an explicit expression for the weak wall BC operator:

$$\begin{aligned} & - \int_{\Gamma_G} \delta p \, \rho (u_i n_i - 0) \, d\Gamma \\ & + \int_{\Gamma_G} \delta u_i n_i \, p \, d\Gamma \\ & - \int_{\Gamma_G} \delta T \, \rho c_v T_g (u_i n_i - 0) \, d\Gamma. \end{aligned} \quad (21)$$

We note that the first two integrals above are the “adjoint” consistency and consistency terms for the pressure, respectively, which were introduced in [25] in an *ad hoc* fashion. In the present work, these are derived using the machinery of weak BCs for advective–diffusive systems. The last integral above is new, and, as evidenced by the numerical results shown in the later sections, presents a key modification that renders the formulation stable and robust.

We now state the complete weak temperature wall BC operator, adding the convective stabilization contributions not covered by the Taylor expansion approach and the contributions from the viscous and heat-flux terms:

$$\begin{aligned}
& - \int_{\Gamma_G} \delta p \rho (u_i n_i - 0) \, d\Gamma \\
& + \int_{\Gamma_G} \delta u_i n_i p \, d\Gamma \\
& - \int_{\Gamma_G} \delta u_i \rho [u_j n_j]_- (u_i - 0_i) \, d\Gamma \\
& - \int_{\Gamma_G} \delta u_i 2\mu u_{(i,j)} n_j \, d\Gamma \\
& - \int_{\Gamma_G} \delta u_i n_i \lambda u_{j,j} \, d\Gamma \\
& - \int_{\Gamma_G} 2\mu \delta u_{(i,j)} n_j (u_i - 0_i) \, d\Gamma \\
& - \int_{\Gamma_G} \lambda \delta u_{i,i} (u_j n_j - 0) \, d\Gamma \\
& + \int_{\Gamma_G} \delta u_i \frac{C_I \mu}{h} (u_i - 0_i) \, d\Gamma \\
& - \int_{\Gamma_G} \delta T \rho c_v T_g (u_i n_i - 0) \, d\Gamma \\
& - \int_{\Gamma_G} \delta T \rho c_v [u_j n_j]_- (T - T_g) \, d\Gamma \\
& - \int_{\Gamma_G} \delta T \kappa T_{,i} n_i \, d\Gamma \\
& - \int_{\Gamma_G} \kappa \delta T_{,i} n_i (T - T_g) \, d\Gamma \\
& + \int_{\Gamma_G} \delta T \frac{C_I \kappa}{h} (T - T_g) \, d\Gamma.
\end{aligned} \tag{22}$$

Here, C_I is a positive constant chosen just large enough to guarantee stability ($C_I = 4$ in this work), h denotes the local mesh size, $(\cdot)_{(i,j)}$ denotes the symmetrization of the tensor $(\cdot)_{i,j}$, and $[\cdot]_-$ denotes the negative part of its argument.

Remark 1 A direct inspection of the convective matrix in Eq. (17) reveals that it has all zero eigenvalues. This means the convective contributions to the weak BC operator are stability-neutral. The eigenvectors of the convective matrix in Eq. (17) may be expressed as

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ -n_2/n_1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ -n_3/n_1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 0 \\ -n_3/n_1 \\ 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}. \tag{23}$$

Note that only four of the above five eigenvectors are linearly independent.

2.3.2 Heat-flux wall BCs

For the heat-flux wall BC, because the temperature is unknown, the vector $\mathbf{G}$ is now given by

$$\mathbf{G} = \begin{bmatrix} p \\ 0 \\ 0 \\ 0 \\ T \end{bmatrix}. \tag{24}$$

The modification in the $\mathbf{G}$ vector, where the prescribed temperature T_g is substituted for the unknown temperature T , results in a slight modification of the product

$$[\mathbf{A}_i^{\text{adv}\setminus p}]_{\mathbf{G}} n_i (\mathbf{Y} - \mathbf{G}) = \begin{bmatrix} \rho (u_i n_i - 0) \\ 0 \\ 0 \\ 0 \\ \rho c_v T (u_i n_i - 0) \end{bmatrix}, \tag{25}$$

while the pressure-flux contributions remain the same. Explicit expressions for the weak BC operator contributions from the convective and pressure-flux terms become

$$\begin{aligned}
& - \int_{\Gamma_G} \delta p \rho (u_i n_i - 0) \, d\Gamma \\
& + \int_{\Gamma_G} \delta u_i n_i p \, d\Gamma \\
& - \int_{\Gamma_G} \delta T \rho c_v T (u_i n_i - 0) \, d\Gamma.
\end{aligned} \tag{26}$$

The complete weak heat-flux wall BC operator, adding the convective stabilization contributions not covered by the Taylor expansion approach, becomes:

$$\begin{aligned}
& - \int_{\Gamma_G} \delta p \rho (u_i n_i - 0) \, d\Gamma \\
& + \int_{\Gamma_G} \delta u_i n_i p \, d\Gamma \\
& - \int_{\Gamma_G} \delta u_i [\rho u_j n_j]_- (u_i - 0_i) \, d\Gamma \\
& - \int_{\Gamma_G} \delta u_i 2\mu u_{(i,j)} n_j \, d\Gamma \\
& - \int_{\Gamma_G} \delta u_i n_i \lambda u_{j,j} \, d\Gamma \\
& - \int_{\Gamma_G} 2\mu \delta u_{(i,j)} n_j (u_i - 0_i) \, d\Gamma \\
& - \int_{\Gamma_G} \lambda \delta u_{i,i} (u_j n_j - 0) \, d\Gamma \\
& + \int_{\Gamma_G} \delta u_i \frac{C_I \mu}{h} (u_i - 0_i) \, d\Gamma \\
& - \int_{\Gamma_G} \delta T \rho c_v T (u_i n_i - 0) \, d\Gamma \\
& - \int_{\Gamma_G} \delta T h_T \, d\Gamma,
\end{aligned} \tag{27}$$

where h_T is a known prescribed heat flux at the wall.

2.4 Numerical procedures

Streamline upwind/Petrov–Galerkin (SUPG) stabilization [1, 26, 31–38] is employed as the primary stabilizing mechanism in this work. For the details of the SUPG operator employed in the present computations, we refer the reader to [25]. Discontinuity-capturing (DC) operator [30, 39–45] is added for additional stability in the presence of shocks. Zeroth Euler–Jacobian of the transformation between the conservation and entropy variables (see [30, Eq. A.82]) is employed in the definition of the DC operator. The method is an extension of the δ_{91} definition developed in [32], where only the convective part of the full residual operator was employed. For the details of the entropy-variable-based DC operator employed in the present computations, we refer the reader to [29, 46].

The semi-discrete compressible-flow equations are integrated in time using a second-order accurate, unconditionally stable, implicit generalized- α method [47–49]. In this work, we utilize the generalized- α parameter of $\rho_\infty = 0.5$. Solution of the nonlinear algebraic equation system resulting after the generalized- α discretization is obtained at each time step using a Newton–Raphson approach. GMRES with block-diagonal preconditioning [50] is used to approximately solve the linear equation system at each nonlinear iteration. The tangent matrix of the Newton–Raphson algorithm is derived analytically and is explicitly computed and stored in a sparse format at each nonlinear iteration. A relative tolerance of 10^{-3} for the residual norm is considered as the convergence criterion for both linear and nonlinear equation solvers.

Remark 2 The formulations presented here are developed on a fixed spatial domain in the Eulerian frame. This was done intentionally so as not to distract the reader from the main focus of the paper on weak BCs for compressible flows. Extension to moving domains using Arbitrary Lagrangian–Eulerian (ALE) or Space–Time (ST) formulations [51] should be relatively straightforward.

Remark 3 To develop the tangent matrix contributions from the weak BC term $-\int_{\Gamma_G} \delta T \rho c_v T (u_i n_i - 0) d\Gamma$ in Eq. (27), we recommend that linearization is performed for

both the temperature and velocity unknowns. The density field may be considered “frozen” during linearization.

3 Results and discussion

In this section, we present the validation of the proposed methodology and demonstrate the effectiveness of the weak wall BCs in capturing accurate flow behavior across subsonic and transonic regimes. We consider 2D test cases of flow over the NACA 0012 and RAE 2822 airfoils and a 3D test case of flow over the ONERA M6 wing at different Reynolds and Mach numbers.

3.1 Flow over the NACA 0012 airfoil

The NACA 0012 airfoil serves as a fundamental validation test case due to its simple symmetric geometry and extensive experimental and numerical-computation data available for comparison [53]. In this study, we select the NACA 0012 airfoil with freestream flow conditions of $M = 0.35$, $Re = 3 \times 10^6$, $T = 300$ K, and $\alpha = 5.88^\circ$, which has been thoroughly investigated experimentally with a comprehensive set of pressure-coefficient data reported in [52].

To assess the convergence of the proposed method employing weak BCs, we construct three unstructured meshes with progressively finer levels of resolution. The computational domain is rectangular, with far-field boundaries located at $100C$, where $C = 1$ is the chord length of the airfoil, to minimize its influence on the near-airfoil field. The airfoil geometry is based on the analytical definition of NACA 0012 provided by the NASA Turbulence Modeling Resource [53]. The computational domain is discretized by extruding a 2D triangular mesh one unit length in the spanwise direction, resulting in a single layer of 3D prismatic elements. To enforce two-dimensionality on the 3D mesh, the governing equations are solved with planar constraints by explicitly setting the spanwise contribution to zero. Each mesh incorporates a size-box refinement extending up to $5C$ from the airfoil and boundary-layer refinements to capture near-wall flow behavior. Additionally, the leading and trailing edges are further refined to properly resolve the curvature at the leading edge and the sharp geometry at the trailing edge. Table 1 summarizes the element sizes and boundary-layer parameters used for the coarse, medium, and fine meshes. Figure 1a shows the medium

Table 1 NACA 0012 airfoil mesh parameters

Mesh	Airfoil surface			Boundary layer			Size box	Outer domain
	Leading/trailing edge	Maximum spacing	Growth rate	Number of layers	First layer height	Growth rate		
Coarse	0.008	0.08	1.05	5	0.008	1.2	0.6	18
Medium	0.004	0.04	1.05	7	0.004	1.2	0.4	16
Fine	0.002	0.02	1.05	9	0.002	1.2	0.2	14

mesh, highlighting the near-airfoil boundary-layer refinements. It is important to note that the first-layer height in our boundary-layer discretization is significantly coarser in the wall-normal direction compared to the standard mesh grid provided by NASA Turbulence Modeling Resource [53]. However, we demonstrate that the proposed weak boundary condition helps attain accurate flow behavior even on these coarser meshes.

To simulate the selected case, the airfoil surface is prescribed a no-slip condition enforced weakly, while the freestream conditions are imposed strongly at the far-field boundaries. The airfoil surface is further assumed to be adiabatic, i.e., with a zero heat-flux BC imposed. The temperature-dependent viscosity, μ , is set according to Sutherland's law as

$$\mu = \mu_r \left(\frac{T}{T_r} \right)^{3/2} \frac{T_r + S}{T + S}, \quad (28)$$

and the thermal conductivity, κ , is calculated using

$$\kappa = \frac{\mu c_p}{Pr} \quad \text{and} \quad c_p = \frac{\gamma R}{\gamma - 1}, \quad (29)$$

where Pr is the Prandtl number and c_p is the specific heat at constant pressure. In this study, $\mu_r = 1.716 \times 10^{-5}$ kg/(m s), $T_r = 273.15$ K, $S = 110.4$ K, $Pr = 0.72$, $R = 287$ J/(kg K), and $\gamma = 1.4$ for air. A time step of 5×10^{-5} s is used for all meshes.

Figure 1b shows the pressure coefficient (C_p) plotted along the airfoil surface for each mesh compared with the experimental results reported in [52]. The results show convergence of the pressure coefficient under mesh refinement, while maintaining good agreement with the experimental data. To demonstrate the advantages of the weak

BC enforcement, the same case is simulated on the medium mesh using strong BC enforcement for comparison. Figure 2 compares the pressure coefficient distribution obtained using the weak and strong BCs on the medium mesh. The results clearly show that the weak BC demonstrates a better agreement with the experimental data, whereas the strong BC deviates significantly. The corresponding Mach number contours and streamlines in Fig. 2 further highlight the difference in the flow behavior between the two methods. With weak BC enforcement, the flow remains predominantly attached along the airfoil surface. In contrast, the strong BC enforcement produces an extensive separated region on the upper surface, leading to vortex formation and affecting the pressure distribution on the airfoil. These findings demonstrate that weak BC enforcement, by allowing some numerical flow slip, can yield globally accurate flow solutions on meshes with insufficient boundary-layer refinement for strong BC enforcement.

3.2 Transonic flow over the RAE 2822 airfoil

We apply the proposed methodology to a more complex airfoil geometry, the RAE 2822 [56], under transonic flow conditions. The RAE 2822 also serves as a standard test case for validating compressible-flow solvers due to the presence of a strong shock on the upper surface, which leads to boundary-layer separation towards the trailing edge. Specifically, we consider the case at $M = 0.729$, $Re = 6.5 \times 10^6$, $T = 255.556$ K, and $\alpha = 2.31^\circ$ which has been extensively studied and experimentally documented in [54]. Similar to the previous NACA 0012 study, three unstructured meshes are constructed using identical computational domain definitions. In this case, however, the airfoil surface discretization is kept consistent with the standard mesh provided by the NPARC Alliance Verification and Validation Archive

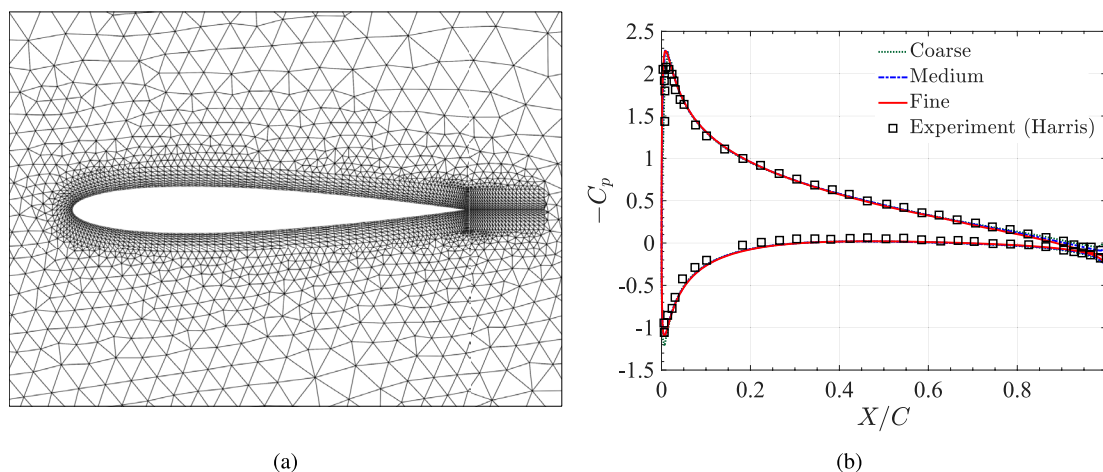


Fig. 1 **a** Near-airfoil medium mesh of the NACA 0012 airfoil. **b** Pressure coefficient computed using three meshes and compared with the experimental data [52] for the case of $M = 0.35$, $Re = 3 \times 10^6$, and $\alpha = 5.88^\circ$

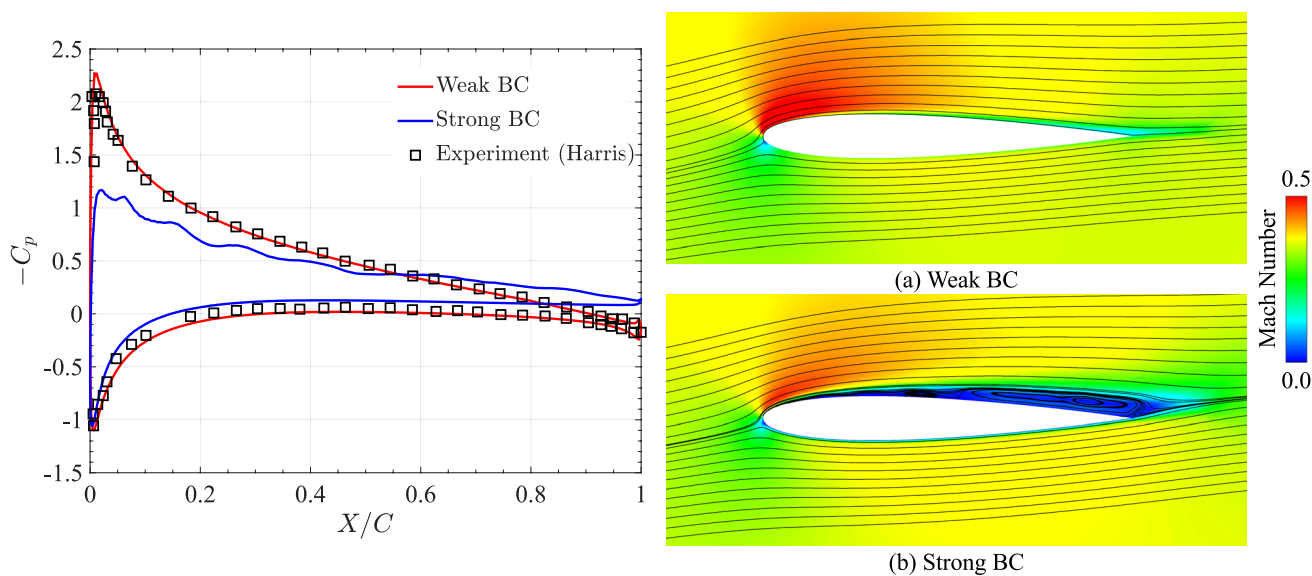


Fig. 2 Pressure coefficient (left) and Mach number superposed with flow streamlines (right) obtained using the medium mesh with weak and strong BCs for the NACA 0012 airfoil at $M = 0.35$, $Re = 3 \times 10^6$, and $\alpha = 5.88^\circ$

Table 2 RAE 2822 airfoil boundary-layer mesh parameters

Mesh	Number of layers	First layer height	Growth rate
Coarse	12	0.002	1.2
Medium	15	0.001	1.2
Fine	18	0.0005	1.2

[56], while the wall-normal spacing for the boundary-layer refinement is progressively relaxed to generate the three mesh levels: coarse, medium, and fine. The boundary-layer parameters for each mesh are summarized in Table 2. All three meshes remain significantly coarser in the wall-normal direction compared to the standard NPARC mesh. On the airfoil surface, a no-slip boundary condition is imposed weakly with the adiabatic wall BC assumption.

The temperature-dependent viscosity and thermal conductivity values are evaluated using Eqs. (28) and (29), with $\mu_r = 1.716 \times 10^{-5}$ kg/(m s), $T_r = 273.15$ K, $S = 110.4$ K, $Pr = 0.72$, $R = 287$ J/(kg K), and $\gamma = 1.4$. A time step of 5×10^{-5} s is applied to all meshes.

Figure 3a shows the pressure coefficient distribution obtained from each mesh and compared with the experimental dataset from [54]. All three meshes capture the shock on the upper surface of the airfoil near $X/C = 0.5$ – 0.6 , which is in close agreement with the experimental results. Moreover, a good convergence is observed as the mesh is refined from coarse to fine. Figure 3b compares the pressure coefficient obtained from the fine mesh with numerical results reported from the WIND and NPARC [55], showing good

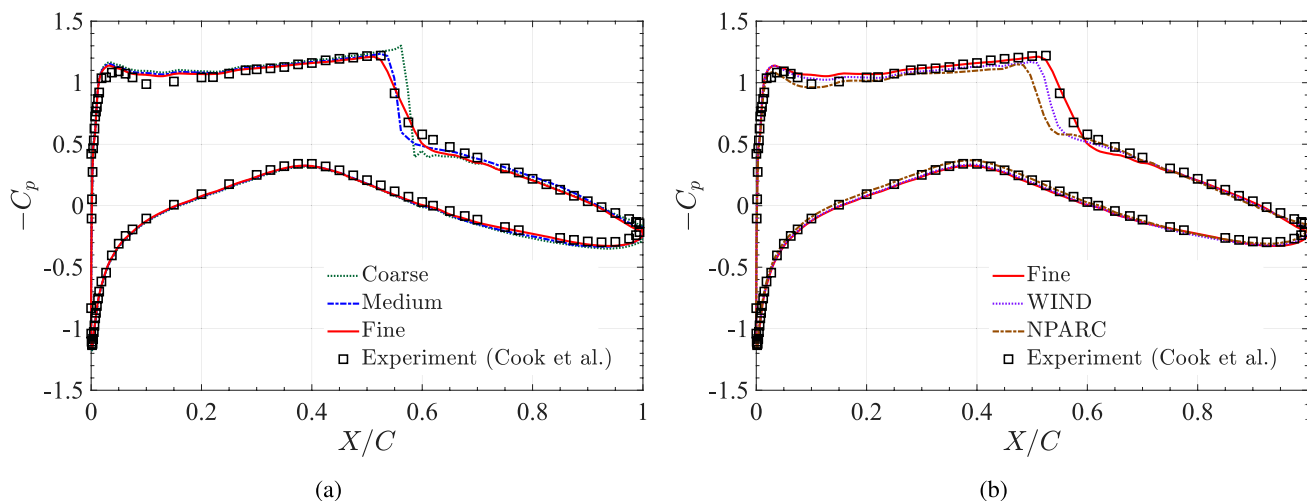


Fig. 3 **a** Pressure coefficient computed using three meshes and compared with experimental data [54] for the RAE 2822 airfoil. **b** Comparison of the pressure coefficient computed on the fine mesh with experimental and simulation results from WIND and NPARC [55]

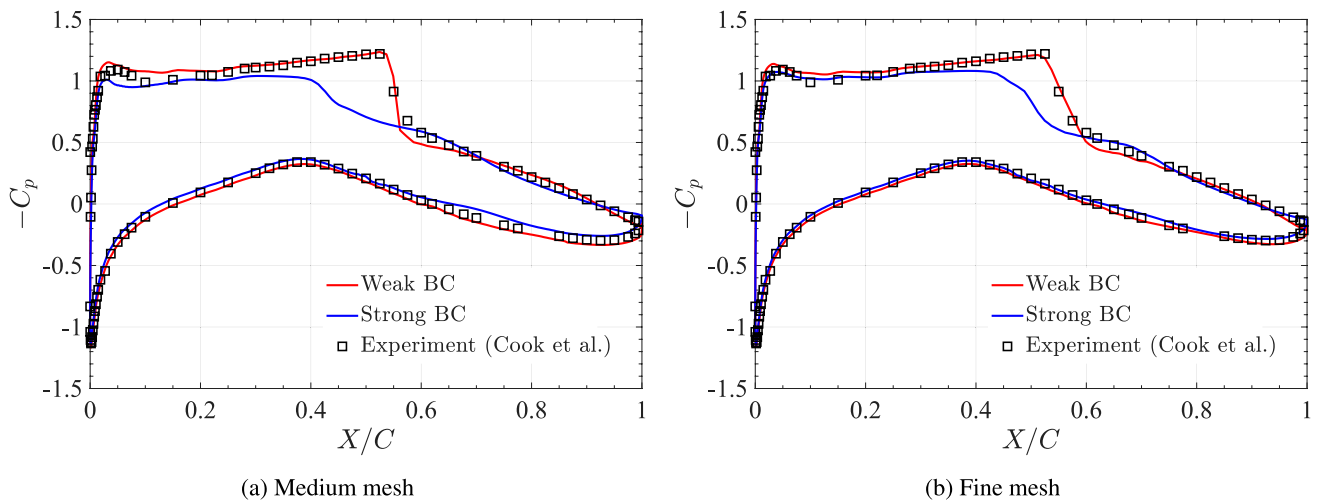


Fig. 4 Comparison of the pressure coefficient obtained using the weak and strong BC formulations for the RAE 2822 airfoil. **a** Medium mesh results; **b** Fine mesh results

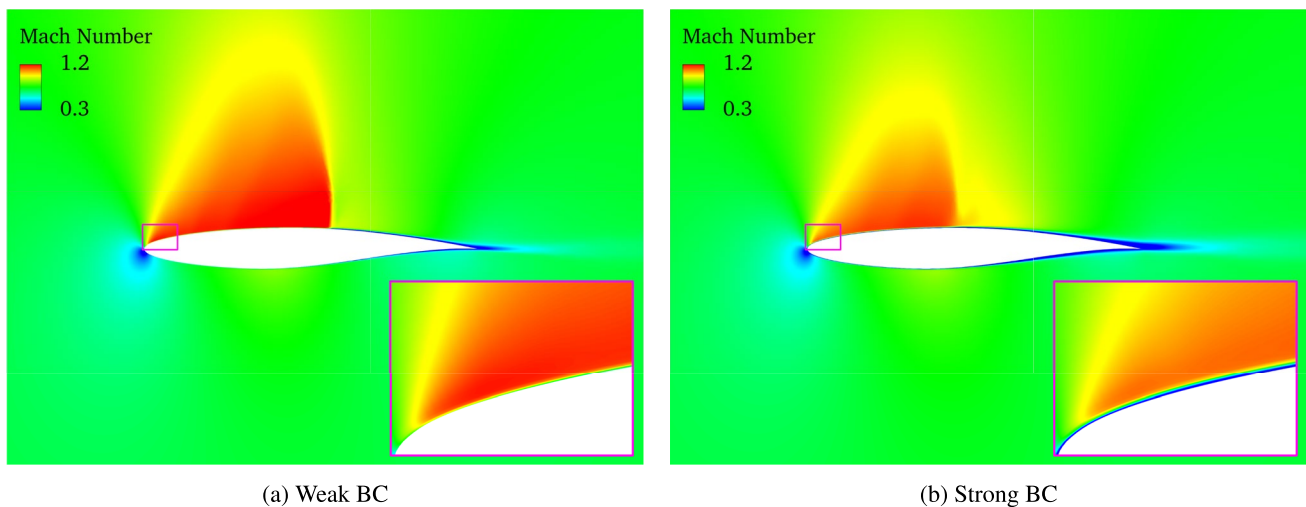


Fig. 5 Mach number distribution obtained using the medium mesh for RAE 2822 with **a** weak BC and **b** strong BC

agreement with both reference datasets. More importantly, the fine mesh in the present study employs a wall-normal element size that is approximately 150 times larger than the standard mesh used in the WIND and NPARC references, yet it still accurately captures the shock location and strength, demonstrating the effectiveness of the weak BC formulation. Moreover, Fig. 4 compares the results obtained using the weak and strong BC enforcement on both medium and fine meshes. The results clearly show that the strong enforcement fails to capture the shock, underscoring the effectiveness of the weak BC approach. (Although the strong BC solution clearly improves with mesh refinement, it is still far from the experimental pressure results in the shock region because the wall-normal mesh resolution is too low to capture the boundary layer.) This behavior is further illustrated in Fig. 5, which compares the Mach number distribution on the medium mesh using the weak and strong

BCs. In the strong BC case, the no-slip condition is fully constrained along the surface, which causes premature flow separation and affects the shock strength. In contrast, the weak BC allows the flow to numerically slip on the surface, enabling a more natural flow separation downstream in the aft region of the airfoil and resulting in a more accurately captured shock.

3.3 ONERA M6 wing

We consider the ONERA M6 wing 2308 configuration, a test case with extensive experimental documentation of the pressure coefficient distribution reported in [57]. The flow is transonic with freestream conditions $M = 0.84$, $Re = 14.6 \times 10^6$ based on the root chord, $T = 300$ K, and $\alpha = 3.06^\circ$. Despite its geometric simplicity, the ONERA M6 wing develops complex flow features at these conditions,

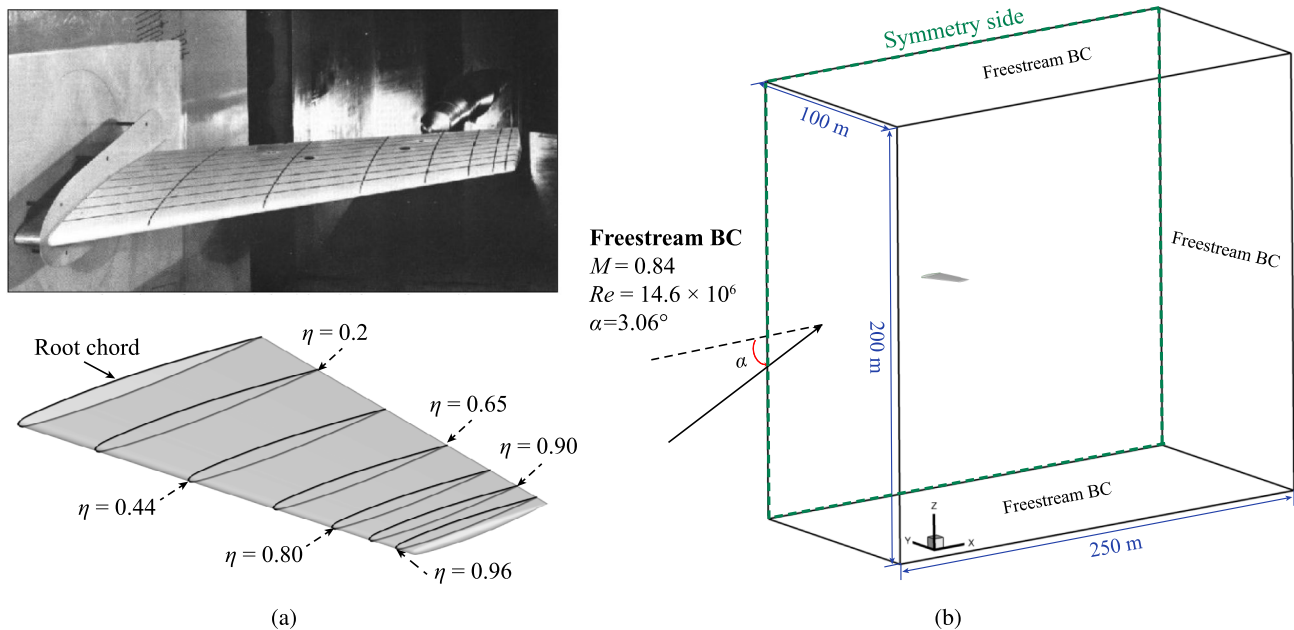


Fig. 6 **a** ONERA M6 wind-tunnel setup [57] and CAD geometry showing six spanwise cuts; **b** Computational domain and BCs

Table 3 ONERA M6 wing mesh parameters

Mesh	Total number of elements	Element size on wing surface (m)	Number of boundary layers	First layer element height (m)	Growth ratio	Outer domain element size (m)
Coarse	1,296,546	0.02	9	0.002	1.2	25
Medium	5,317,478	0.01	12	0.001	1.2	12.5
Fine	13,558,505	0.005	15	0.0005	1.2	6.25

such as flow separation, shock boundary-layer interactions, and spanwise variation of the shock location. The experimental arrangement and the CAD geometry used here and shown in Fig. 6a are taken from the NASA Turbulence Modeling Resource [60] and the NPARC Alliance Verification and Validation Archive [61]. Six different spanwise locations (η) are taken on the wing surface to report the pressure coefficient profiles. Figure 6b shows the computational domain and boundary conditions for this transonic case. A symmetry boundary condition is applied on the plane emanating from the wing root, while the remaining outer faces are subject to freestream BCs. On the wing surface, a no-slip boundary condition is enforced weakly along with an adiabatic condition (zero heat flux). Thermophysical properties are calculated according to Eqs. (28) and (29) with $\mu_r = 1.716 \times 10^{-5}$ kg/(m s), $T_r = 273.15$ K, $S = 110.4$ K, $Pr = 0.72$, $R = 287$ J/(kg K), and $\gamma = 1.4$. A time step size of 5×10^{-5} s is used in this analysis.

We construct three unstructured meshes using tetrahedral elements with an increasing level of refinement to show convergence of the pressure coefficient. Table 3 presents the summary of the mesh statistics for the coarse, medium, and

fine meshes. Figure 7 shows the medium mesh with boundary layer and size box refinements to capture the complex flow separation and shock formation on the wing. The pressure coefficient distribution at different spanwise locations for three meshes is shown in Fig. 8. It is evident that mesh refinement has a minor effect near the smooth flow region away from the shocks, but it plays a significant role in accurately capturing the sharper shock features. The medium and fine meshes yield nearly identical pressure coefficient distributions across most of the wing surface, closely matching the experiments [57], and indicating mesh convergence with minimal sensitivity to further mesh refinements. Furthermore, Fig. 9 compares the pressure coefficient obtained with our fine mesh solution and the simulation results of FUN3D (SA-neg) [58] and WIND [59] together with the experimental data [57]. Overall, the result from the present simulation is in good agreement with both reference numerical results that were obtained using a much finer mesh. More specifically, the first-layer element height normal to the wing surface for the fine mesh is 0.0005 m, which is roughly 30 times larger than that employed in FUN3D and WIND simulations. Despite the much coarser boundary-layer mesh, our

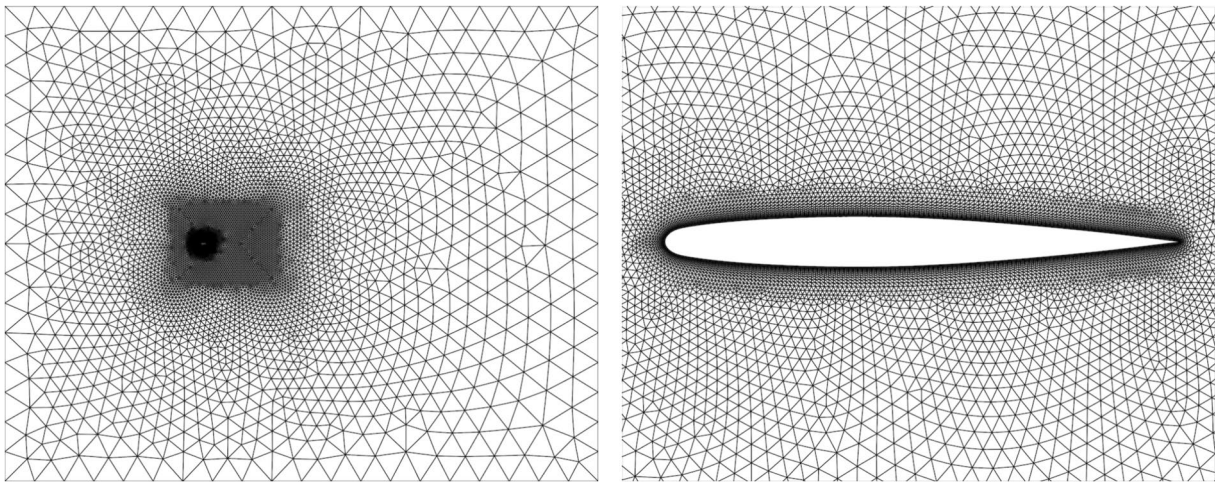


Fig. 7 ONERA M6 wing mesh on a spanwise cut. Full domain (left) and zoom on the wing surface (right)

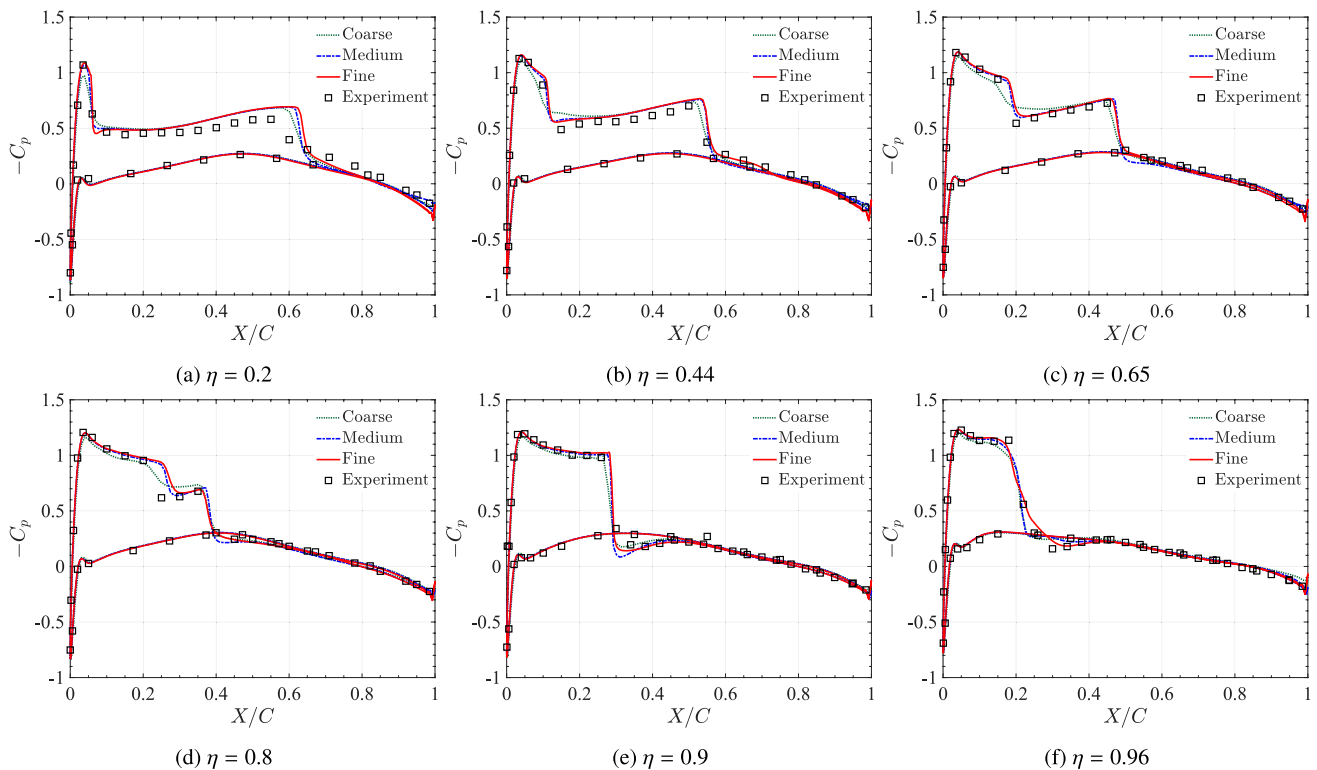


Fig. 8 ONERA M6 wing pressure coefficient at six spanwise locations computed on the three meshes and compared with experimental data from [57]. See Fig. 6 for the spanwise-cut locations denoted by η

simulations of this 3D case confirm that weak BCs enable accurate near-wall predictions without requiring highly refined boundary-layer meshes.

To further highlight the impact of weak BCs, the fine-mesh case is recomputed using strongly imposed no-slip BCs at the wing surface. Figure 10 compares the resulting pressure coefficient distribution. The weak enforcement solution matches the experiments more closely than the strong enforcement solution, particularly in predicting the

upper surface suction peak, the shock position, and the post-shock pressure recovery. Moreover, Fig. 11 compares pressure coefficient contours and shock structures on the wing obtained with weak versus strong BC enforcement. The weak enforcement produces a sharper, more coherent shock along the span, whereas the strong enforcement results in a more diffused surface pressure field and the absence of well-defined shocks.

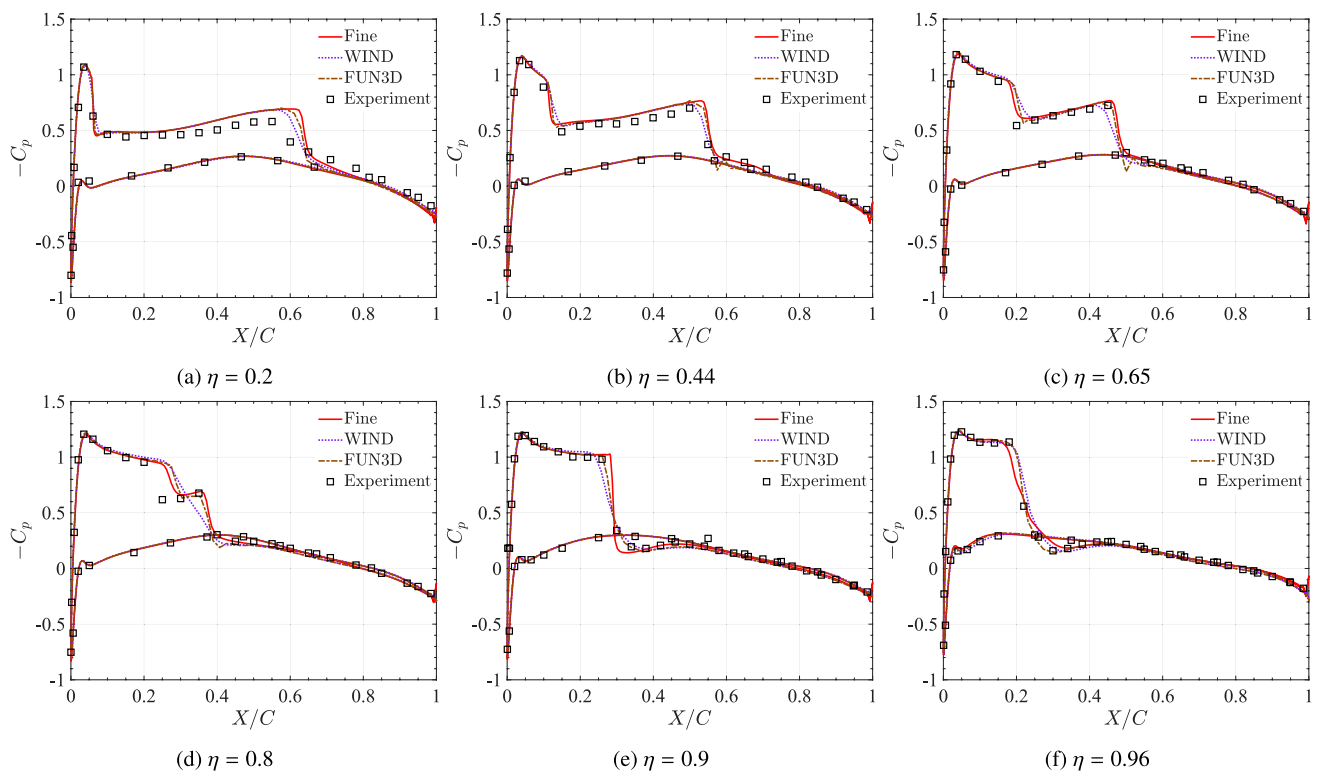


Fig. 9 ONERA M6 wing pressure coefficient at six spanwise locations computed on the fine mesh and compared with experimental data from [57] and simulation results from the FUN3D [58] and WIND [59] codes

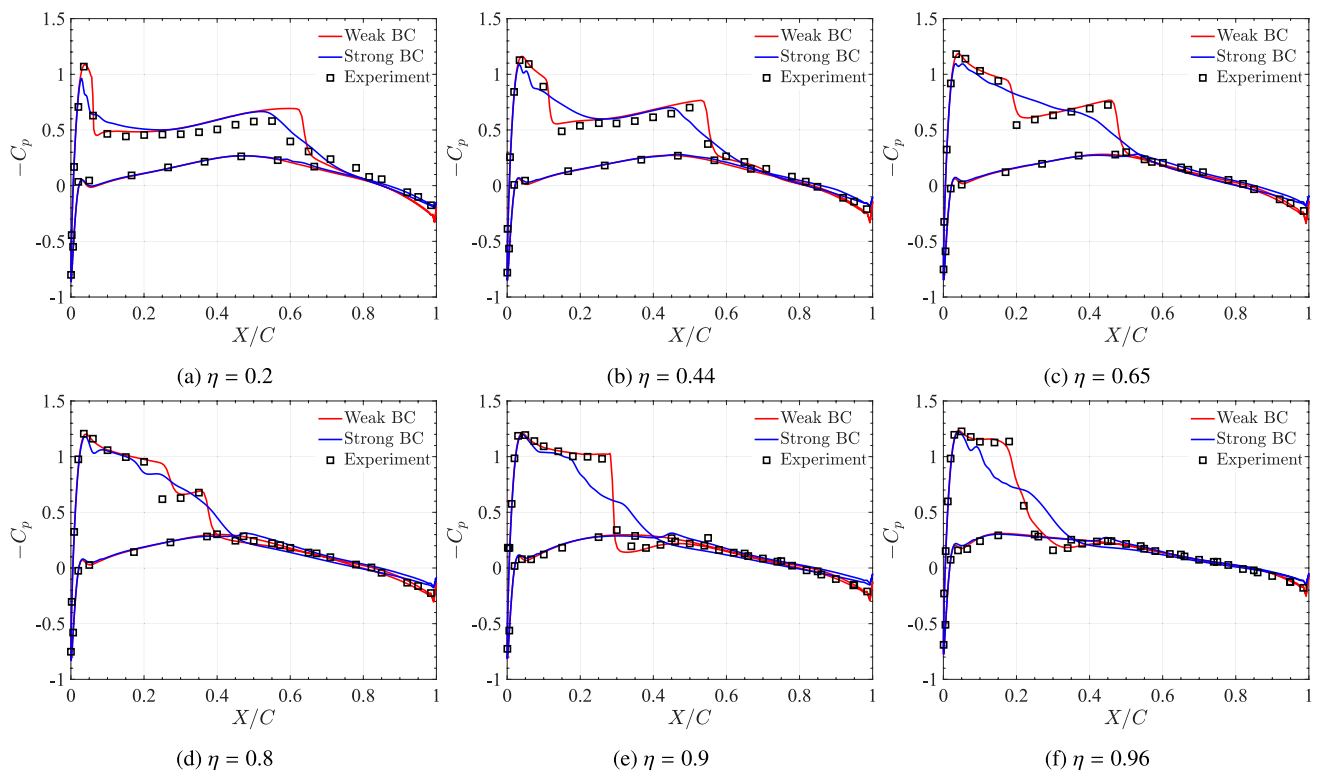


Fig. 10 ONERA M6 wing pressure coefficient at six spanwise locations computed on the fine mesh. Comparison of the weak and strong BC imposition

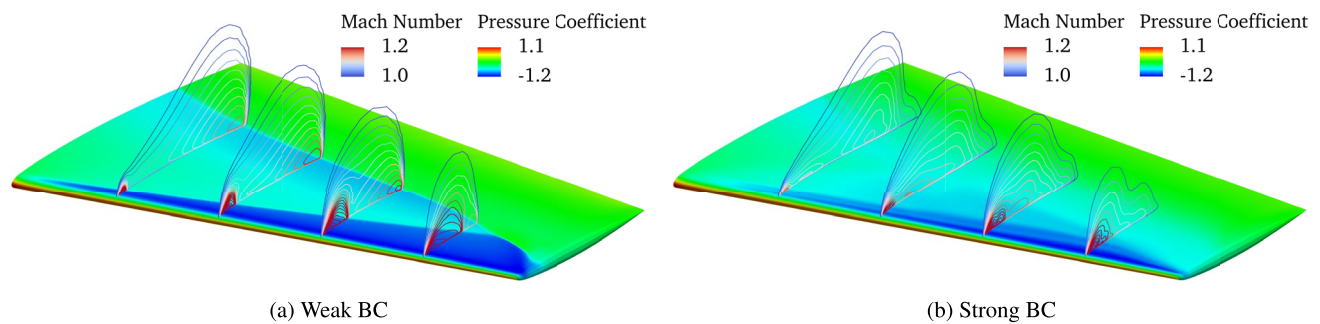


Fig. 11 Surface pressure coefficient and Mach number contours for **a** weak BC solution and **b** strong BC solution, both computed on the fine mesh for ONERA M6 wing

The simulated pressure distributions exhibit very good agreement with the reference data across the span, with the weak BC consistently outperforming the strongly enforced variant in matching the measured suction peaks, shock locations, and downstream pressure recovery. In addition, the accurate location of shocks and the associated pressure drops at different spanwise locations highlight the robustness of the weak BC formulation. We also note that the choice of entropy-variable-based DC operator also factors into the accuracy of the computational results for this 3D transonic flow.

4 Conclusions and outlook

Using a framework of advective–diffusive systems, we carefully constructed weak BC operator contributions arising from the advective and pressure fluxes of the compressible-flow equations. We considered both temperature and heat-flux boundary conditions that are commonly set at solid walls. Through this construction, we identified terms that were not considered previously, whose absence likely explains the underwhelming performance of weak BCs in prior efforts. Specifically, previous works required strong enforcement of the temperature BCs in the hypersonic regime [28] and strong enforcement of the normal velocity component with heat-flux BCs [29] to prevent numerical instabilities. The newly derived terms address these challenges, ensuring the accuracy and robustness of a fully weak BC framework for compressible flows.

The stability and accuracy of the proposed formulations are demonstrated through several challenging 2D and 3D test cases. Aerodynamic simulations were conducted for a range of benchmark configurations, including 2D subsonic flow over the NACA 0012 airfoil, 2D transonic flow over the RAE 2822 airfoil, and 3D transonic flow over the ONERA M6 wing. The predicted pressure distributions were in good agreement with the reference experimental and numerical data. More importantly, accurate results were obtained

using significantly coarser boundary-layer meshes than the standard grids used in benchmark computations [53, 56]. This demonstrates the effectiveness of the weakly enforced essential BCs in capturing wall-bounded compressible-flow behavior without the need for excessive refinements near the solid surfaces.

The weak BC techniques, as presented here, are unique to the numerical methods that emanate from weak or variational formulations in fluid mechanics. To the best of the authors' knowledge, no direct equivalents of weak BCs exist in finite-difference or finite-volume formulations. The closest analogs of weak BCs in traditional computational fluid dynamics (CFD) are the near-wall modeling approaches [62], which are employed as part of Large Eddy Simulation (LES) and Reynolds-Averaged Navier–Stokes (RANS) models of turbulent flows [63]. In near-wall modeling, the no-slip boundary condition is replaced by a traction boundary condition, where the traction magnitude is determined from a chosen “law of the wall” [64], also leading to a numerical slip. We believe that adopting a weak BC formulation to a more classical CFD framework could have a significant impact on the larger computational fluids community and can unify ideas around near-wall models. Some early developments in this direction can be found in [12].

Moreover, weak BCs provide a natural framework for immersed methods, such as IMGA and SBM, where they have been developed and applied to incompressible flow to produce accurate and stable solutions on non-boundary-fitted meshes. The proposed weak BC formulations can further extend the capabilities of immersed methods to compressible flow, enabling reliable simulations over complex geometries. This development not only expands the use of immersed methods into new flow regimes but also lays the groundwork for robust multiphysics analysis involving compressible flows [46, 65].

All the examples computed in the present paper involve some form of shock/boundary layer interaction (SBLI). SBLI is notoriously challenging, and the computational results shown highlight the fact that under-resolved and/

or improperly modeled boundary layers (represented here through strong BC solutions on coarse meshes) can lead to a significant loss of accuracy in the prediction of the shock location and strength. Our weak BC formulation, by allowing some numerical flow slip, appears to overcome this issue: Even on very coarse meshes, the shocks are captured with an impressive degree of accuracy. This makes weak BCs a practical approach for the simulation of industrial-scale wall-bounded compressible flows and also a good platform for computational-technology development for turbulent SBLL.

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Author contributions MJ Implemented the methodology, ran the test cases, prepared the figures, and wrote parts of the manuscript; MR Developed earlier versions of the code, ran preliminary tests, and commented on the manuscript; MRI ran test cases, processed the data and reviewed the manuscript; M-CH Conceptualized the research, provided advising, secured funding, wrote parts of the manuscript, and reviewed and commented on the manuscript; YB Developed the methodology, provided advising, wrote parts of the manuscripts, commented on the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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